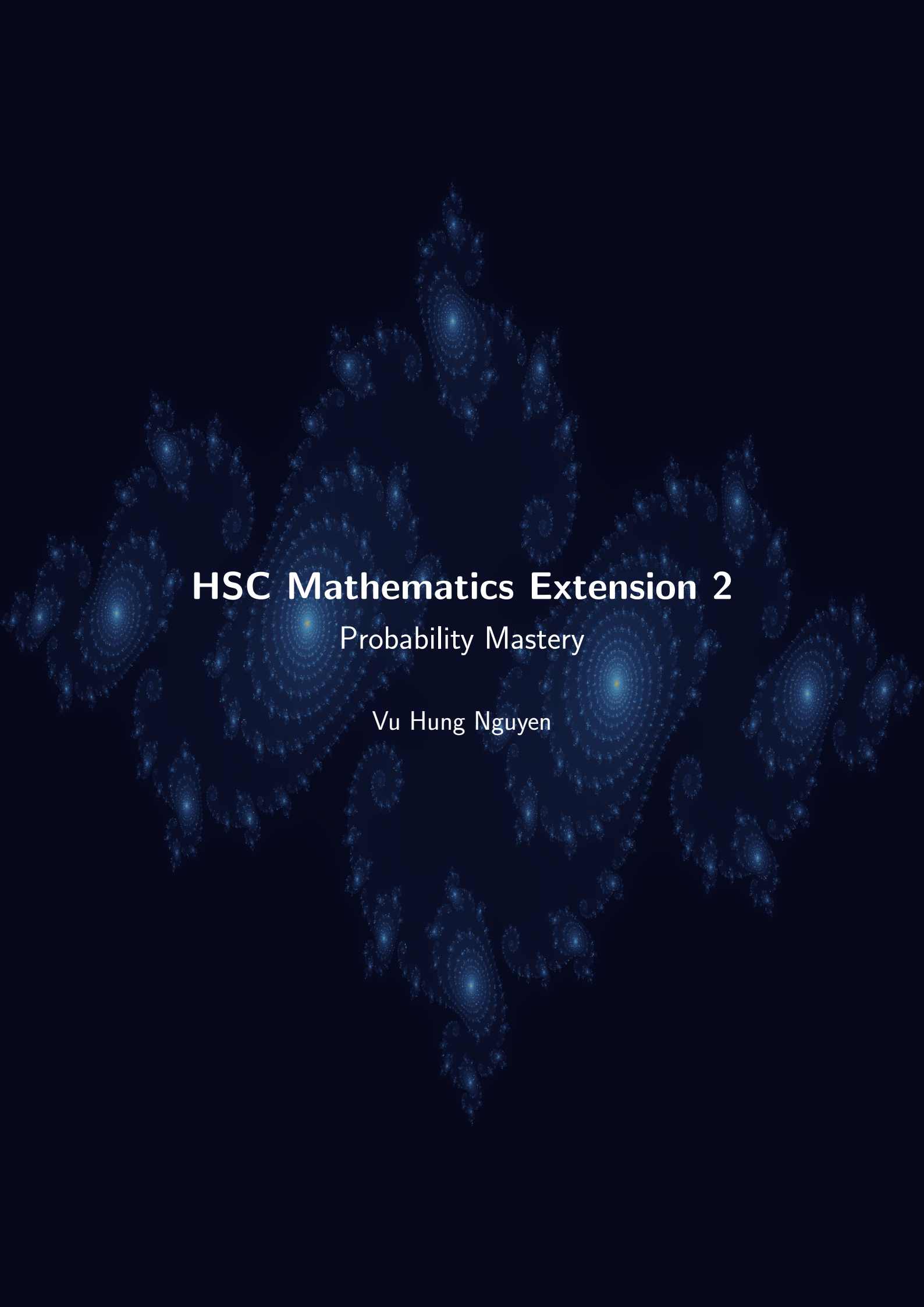




HSC Mathematics Extension 2

Probability Mastery

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Download PDF and resources:

<https://github.com/vuhung16au/math-olympiad-ml>

“Pure mathematics is, in its way, the poetry of logical ideas.”

Albert Einstein

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1 Introduction

1.1 Project Overview

This booklet is a focused probability companion for NSW HSC Mathematics Extension 1 students. It combines the main ideas from probability and discrete probability distributions into one coherent practice resource. The goal is to help students move smoothly from event-based reasoning to random variables, expectation, and binomial models.

1.2 Target Audience

This resource is written for Extension 1 students who want more structured probability practice than a standard textbook chapter usually offers. It is also suitable for tutors and teachers who want a compact booklet with model solutions in one place.

1.3 How to Use This Booklet

- Read the fundamentals review first to refresh notation, rules, and common traps.
- Work through Part 1 carefully because the detailed solutions model the standard reasoning expected in HSC-style written responses.
- Use Part 2 for repetition and fluency. Try each problem before turning the page to read the upside-down hint.
- Pay attention to wording such as *at least*, *exactly*, *given that*, and *independent*, since these phrases determine the method.
- Keep checking whether a problem is about events, counting, or a random variable. Many errors come from mixing these levels.

1.4 Topics Covered

The booklet develops the following Extension 1 ideas:

- sample spaces and equally likely outcomes
- events, complements, unions, and intersections
- addition and multiplication rules
- conditional probability and independence
- counting-based probability links
- discrete random variables and probability tables
- expected value and variance
- binomial distributions and repeated trials

1.5 Where This Leads Next

This booklet is written for HSC Mathematics Extension 1, but it also prepares students for important ideas that appear later in HSC Mathematics Extension 2.

Extension 2, *Continuous probability distributions*, extends the random-variable ideas in this booklet from discrete tables to continuous models, where probabilities are found from densities and areas rather than separate point values.

Extension 2, *Binomial distributions*, builds further on the repeated-trial and expectation ideas developed here. In Extension 1, students learn to recognise binomial structure, calculate probabilities, and interpret mean and variance. Extension 2 then pushes these ideas into a more formal and more powerful setting.

So this booklet can be read not only as Extension 1 revision, but also as part of the conceptual bridge into later probability study.

2 Fundamentals Review

This section collects the main rules and interpretations used throughout the booklet. It is not meant to replace understanding; it is meant to make that understanding easier to apply under exam pressure.

2.1 Core Language

A **sample space** is the set of all possible outcomes of an experiment. An **event** is a subset of that sample space.

Example. If a fair die is rolled, then

$$S = \{1, 2, 3, 4, 5, 6\}.$$

The event “an even number is rolled” is

$$E = \{2, 4, 6\}.$$

When outcomes are equally likely,

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{number of possible outcomes}}.$$

2.2 Set Language Refresher

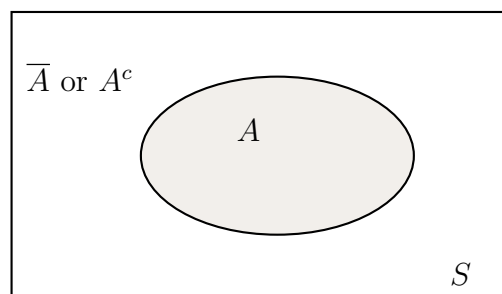
Probability uses the language of sets, so it is worth recalling a few basic terms.

Subsets. If every element of a set A also belongs to a set B , then A is a **subset** of B , written

$$A \subseteq B.$$

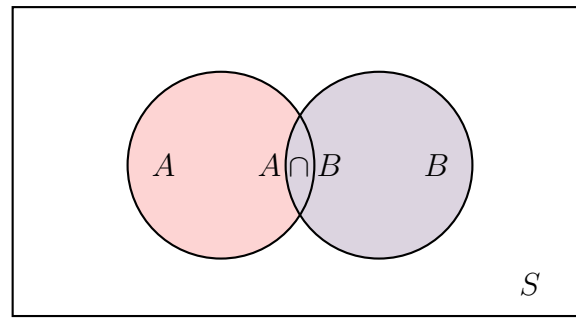
In probability, an event is always a subset of the sample space.

The universal set and complement. The **universal set** is the set containing all objects under discussion. In probability, this role is usually played by the sample space S . If A is a set, then its **complement** is the set of elements in the universal set that are not in A . It may be written as $\bar{A}$ or A^c .



Intersection and union. For two sets A and B :

- $A \cap B$ is the **intersection**, meaning the elements common to both sets
- $A \cup B$ is the **union**, meaning the elements in A or B or both



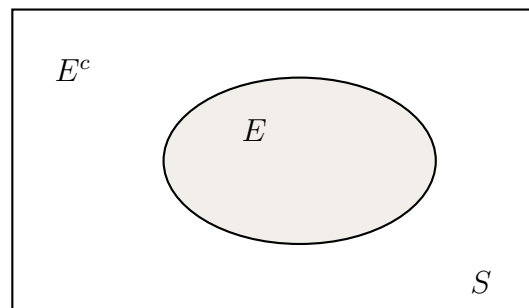
In a Venn diagram, the union is the whole region covered by the two circles, while the intersection is the overlap in the middle.

2.3 Complements, Unions, and Intersections

If A is an event, then its complement means “not A ”. It may be written as A^c or $\bar{A}$.

$$P(A^c) = 1 - P(A).$$

Complementary events and the word “not”. In probability questions, the word “not” often signals a complement. If E is an event, then the complement of E is the event “ E does not occur”. It is written as E^c or sometimes $\bar{E}$.



In the diagram, the rectangle is the sample space S . The oval is the event E . Everything *outside* the oval but still inside the rectangle represents E^c , meaning “not E ”.

Example. If E is “rolling a 6 on a fair die”, then E^c is “not rolling a 6”, so

$$P(E^c) = 1 - P(E) = 1 - \frac{1}{6} = \frac{5}{6}.$$

For two events A and B :

- $A \cup B$ means “ A or B or both”
- $A \cap B$ means “ A and B ”

The addition rule is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

The subtraction avoids double-counting the outcomes common to both events.

2.4 Mutually Exclusive Events

Events are **mutually exclusive** if they cannot happen together. Then

$$P(A \cap B) = 0,$$

so the addition rule simplifies to

$$P(A \cup B) = P(A) + P(B).$$

Warning. Mutually exclusive does **not** mean independent. In fact, non-trivial mutually exclusive events are usually dependent.

2.5 Multiplication Rule and Conditional Probability

Conditional probability means the probability of an event given that another event has already happened:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

Rearranging gives the multiplication rule:

$$P(A \cap B) = P(B) P(A | B) = P(A) P(B | A).$$

This is especially useful for selection without replacement.

2.6 Independence

Events A and B are **independent** if knowing that one occurred does not change the probability of the other. Algebraically,

$$P(A \cap B) = P(A)P(B).$$

Equivalently, if $P(B) > 0$,

$$P(A | B) = P(A).$$

Common trap. “Independent” is about probabilities staying unchanged. It is not the same as “different” or “mutually exclusive”.

2.7 Counting Links

Some probability questions are easiest when written as counting problems. Common tools include:

- permutations when order matters
- combinations when order does not matter
- complementary counting when “at least one” is awkward

Example. The probability that at least one head appears in three fair coin tosses is

$$1 - P(\text{no heads}) = 1 - \left(\frac{1}{2}\right)^3 = \frac{7}{8}.$$

If a question involves selecting objects, it is often worth translating the numerator and denominator into combinations. For example, if 2 cards are chosen from a deck and we want both cards to be aces, then

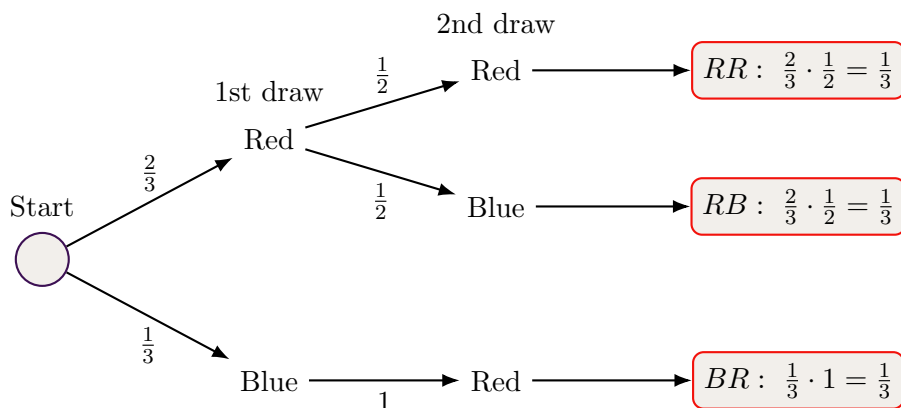
$$P(\text{both aces}) = \frac{\binom{4}{2}}{\binom{52}{2}}.$$

This method is especially clean when order does not matter.

2.8 Tree Diagrams and Staged Experiments

Tree diagrams are useful when an experiment happens in stages. Each branch shows a probability, and the probability of a complete path is found by multiplying along the path.

Example. Suppose a bag contains 2 red counters and 1 blue counter, and two counters are drawn without replacement. The tree diagram is:



The path probabilities add to 1, which is a good quick check.

2.9 Two-Way Tables

Two-way tables are helpful when events are described by two classifications at once, such as subject choices, gender, or transport method.

	B	B^c	Total
A	$n(A \cap B)$	$n(A \cap B^c)$	$n(A)$
A^c	$n(A^c \cap B)$	$n(A^c \cap B^c)$	$n(A^c)$
Total	$n(B)$	$n(B^c)$	$n(S)$

Conditional probabilities then become quick to read. For example,

$$P(A | B) = \frac{n(A \cap B)}{n(B)}.$$

2.10 Discrete Random Variables

A **discrete random variable** takes specific numerical values, each with an associated probability. The probabilities must satisfy:

$$0 \leq P(X = x) \leq 1 \quad \text{and} \quad \sum P(X = x) = 1.$$

These are often shown in a table.

x	$P(X = x)$	Interpretation
0	0.2	no successes
1	0.5	one success
2	0.3	two successes

Bridge note. In this booklet, the random variable takes separate values such as 0, 1, 2, ..., so we can list probabilities in a table. In continuous probability, the possible values fill intervals instead of isolated points, so the table is replaced by a density curve and sums are replaced by integrals. The central ideas stay the same: we still describe how likely values are, and we still study averages and spread.

2.11 Expected Value and Variance

The expected value of a discrete random variable is its weighted average:

$$E(X) = \sum xP(X = x).$$

The variance measures spread:

$$\text{Var}(X) = E(X^2) - [E(X)]^2, \quad \text{where } E(X^2) = \sum x^2P(X = x).$$

The standard deviation is

$$\sigma = \sqrt{\text{Var}(X)}.$$

Interpretation note. The expected value is not necessarily an outcome that must actually occur. It is the long-run average value.

Transition remark. This same story continues in continuous probability. Expected value still represents a long-run average, and variance still measures spread, but the formulas move from sums such as $\sum xP(X = x)$ to integrals over a probability density function. So continuous random variables are best thought of as the next version of the same ideas, not a completely different topic.

2.12 Enrichment Note: A First Glimpse of Measure Theory

Syllabus note. This subsection is enrichment only. Measure theory is **outside** the HSC Mathematics Extension 1 and Extension 2 syllabuses.

At a very high level, measure theory gives a systematic way to assign “size” to sets. In elementary probability, that “size” becomes probability after normalising so that the whole sample space has total size 1.

For a **discrete** random variable with values $x = 0, 1, 2, \dots$, the measurement is a sum of point masses:

$$P(X \in A) = \sum_{x \in A} P(X = x).$$

For a single **continuous** random variable X , the measurement is described by a density function, and probabilities are found by integration:

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

For **two variables** X and Y , probability is measured over regions in the plane. This is why some geometric probability questions can be solved using areas:

$$P((X, Y) \in R) = \iint_R f(x, y) dA.$$

The same idea extends to higher dimensions $\mathbb{R}^n$, where probability is measured over more complicated regions. So the discrete, one-variable continuous, and area-based probability ideas in this booklet can all be viewed as early examples of one larger measuring framework.

2.13 A Worked Variance Example

Suppose

$$P(X = 0) = 0.25, \quad P(X = 1) = 0.50, \quad P(X = 2) = 0.25.$$

Then

$$E(X) = 0(0.25) + 1(0.50) + 2(0.25) = 1.$$

Also,

$$E(X^2) = 0^2(0.25) + 1^2(0.50) + 2^2(0.25) = 1.5.$$

So

$$\text{Var}(X) = 1.5 - 1^2 = 0.5.$$

This example shows that a random variable can have average value 1 while still spreading around that centre.

2.14 Binomial Distribution

A random variable X has a **binomial distribution** when:

- there are a fixed number n of trials
- each trial has two outcomes, success or failure
- the probability of success is constant, say p
- the trials are independent

Then

$$P(X = r) = \binom{n}{r} p^r (1 - p)^{n-r} \quad \text{for } r = 0, 1, 2, \dots, n.$$

Also,

$$E(X) = np, \quad \text{Var}(X) = np(1 - p).$$

Worked example. If a biased coin has probability 0.6 of landing heads and is tossed 4 times, then the probability of exactly 3 heads is

$$\binom{4}{3} (0.6)^3 (0.4) = 4(0.216)(0.4) = 0.3456.$$

The expected number of heads is

$$E(X) = np = 4(0.6) = 2.4.$$

2.15 Sample Proportion as a Random Variable

Suppose a Bernoulli trial has probability of success p , and it is repeated n times independently. If X is the number of successes, then

$$X \sim \text{Bin}(n, p).$$

The **sample proportion** of successes is

$$\hat{p} = \frac{X}{n}.$$

This is also a random variable. Its mean is

$$E(\hat{p}) = p,$$

and its variance is

$$\text{Var}(\hat{p}) = \frac{p(1 - p)}{n}.$$

Interpretation note. The sample proportion fluctuates from sample to sample, but on average it centres around the true population proportion p .

2.16 Normal Approximation to the Binomial

When n is large and the binomial distribution is not too skewed, a binomial random variable can often be approximated by a normal distribution.

If

$$X \sim \text{Bin}(n, p),$$

then a standard approximation is

$$X \approx N(np, np(1-p)).$$

Equivalently, for the sample proportion,

$$\hat{p} \approx N\left(p, \frac{p(1-p)}{n}\right).$$

This lets us use standard normal methods to estimate probabilities that would otherwise require many separate binomial terms.

Continuity-correction note. When approximating a discrete variable such as X by a continuous normal model, it is often better to shift endpoints by 0.5. For example,

$$P(55 \leq X \leq 65)$$

is often approximated by

$$P(54.5 \leq Y \leq 65.5),$$

where Y is the corresponding normal random variable.

Validity warning. A normal approximation may be poor if the sample size is small or if the success probability is very close to 0 or 1. In that case the binomial distribution can be too skewed for the normal curve to be reliable.

2.17 Common Language Triggers

The wording of a probability question often suggests the method:

- **at least one:** usually try a complement
- **given that:** conditional probability
- **independent:** check whether probabilities stay unchanged
- **exactly:** count one precise case
- **at most:** add cases up to a bound or subtract a complement
- **without replacement:** probabilities usually change from stage to stage

2.18 Method Selection Checklist

Before calculating, ask:

- Is this a plain event probability, a conditional probability, or a random-variable question?
- Are the outcomes equally likely?
- Is replacement involved?
- Would a complement be quicker?
- Is this genuinely binomial, or do the trial conditions fail?

3 Part 1: Problems and Solutions (Detailed)

Part 1 contains fully worked model solutions. The emphasis is on method selection, clean notation, and explaining why each rule is appropriate.

3.1 Basic Probability Problems

Problem 3.1: Two dice and a sum of at least 10

Two fair six-sided dice are rolled. Find the probability that the sum is at least 10.

Solution 3.1

There are 36 equally likely ordered outcomes.

The sums at least 10 are:

10 : (4, 6), (5, 5), (6, 4),

11 : (5, 6), (6, 5),

12 : (6, 6).

So there are 6 favourable outcomes.

Hence

$$P(\text{sum at least 10}) = \frac{6}{36} = \frac{1}{6}.$$

Takeaways 3.1

- Start by checking whether outcomes are equally likely.
- With dice, ordered pairs are usually the cleanest sample space.
- “At least” often means grouping several cases together.

Problem 3.2: A card that is a heart or a face card

A card is chosen at random from a standard deck of 52 cards. Find the probability that the card is a heart or a face card.

Solution 3.2

Let

$$H = \{\text{heart}\}, \quad F = \{\text{face card}\}.$$

There are 13 hearts and 12 face cards. The overlap consists of the heart face cards:

$$J\heartsuit, Q\heartsuit, K\heartsuit,$$

so

$$n(H \cap F) = 3.$$

By the addition rule,

$$P(H \cup F) = \frac{13 + 12 - 3}{52} = \frac{22}{52} = \frac{11}{26}.$$

Takeaways 3.2

- Use the addition rule when two descriptions can overlap.
- Subtract the intersection once to avoid double-counting.
- Drawing a quick Venn diagram is often helpful.

Problem 3.3: At least one blue marble

A bag contains 4 red marbles and 3 blue marbles. Two marbles are drawn without replacement. Find the probability that at least one marble is blue.

Solution 3.3

The complement of “at least one blue” is “no blue”, which means both marbles are red. So

$$P(\text{both red}) = \frac{4}{7} \cdot \frac{3}{6} = \frac{12}{42} = \frac{2}{7}.$$

Therefore,

$$P(\text{at least one blue}) = 1 - \frac{2}{7} = \frac{5}{7}.$$

Takeaways 3.3

- “At least one” is often easiest through a complement.
- Without replacement, multiply changing fractions.
- The complement method reduces casework.

Problem 3.4: Mutually exclusive or independent?

One card is drawn from a standard deck. Let

$$A = \{\text{the card is an ace}\}, \quad B = \{\text{the card is a spade}\}.$$

Find $P(A \cap B)$ and decide whether A and B are independent.

Solution 3.4

The only card that is both an ace and a spade is the ace of spades, so

$$P(A \cap B) = \frac{1}{52}.$$

Also,

$$P(A) = \frac{4}{52} = \frac{1}{13}, \quad P(B) = \frac{13}{52} = \frac{1}{4}.$$

Thus

$$P(A)P(B) = \frac{1}{13} \cdot \frac{1}{4} = \frac{1}{52}.$$

Since

$$P(A \cap B) = P(A)P(B),$$

the events are independent.

Takeaways 3.4

- Events can overlap and still be independent.
- Independence is a probability test, not a visual one.
- A single shared outcome does not automatically mean dependence.

3.2 Medium Probability Problems**Problem 3.5: Conditional probability from a class survey**

In a class, 18 students study Mathematics Extension 1, 14 study Chemistry, and 8 study both subjects. One student is selected at random from the class, knowing that the student studies Chemistry. Find the probability that the student also studies Mathematics Extension 1.

Solution 3.5

Let

$$M = \{\text{studies Extension 1}\}, \quad C = \{\text{studies Chemistry}\}.$$

We are asked to find

$$P(M | C).$$

Using conditional probability,

$$P(M | C) = \frac{P(M \cap C)}{P(C)}.$$

Because every student is equally likely to be chosen, we may use counts:

$$P(M | C) = \frac{8}{14} = \frac{4}{7}.$$

Takeaways 3.5

- “Given that” signals conditional probability.
- Once the condition is imposed, the reduced sample space becomes the relevant denominator.
- Counts often make conditional probability faster than decimal probabilities.

Problem 3.6: Checking independence

A fair coin is tossed twice. Let

$$A = \{\text{exactly one head}\}, \quad B = \{\text{the first toss is a head}\}.$$

Determine whether A and B are independent.

Solution 3.6

The sample space is

$$\{HH, HT, TH, TT\}.$$

Event A is

$$A = \{HT, TH\},$$

so

$$P(A) = \frac{2}{4} = \frac{1}{2}.$$

Event B is

$$B = \{HH, HT\},$$

so

$$P(B) = \frac{2}{4} = \frac{1}{2}.$$

The intersection is

$$A \cap B = \{HT\},$$

thus

$$P(A \cap B) = \frac{1}{4}.$$

Now

$$P(A)P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Since

$$P(A \cap B) = P(A)P(B),$$

the events are independent.

Takeaways 3.6

- Independence is tested by comparing $P(A \cap B)$ with $P(A)P(B)$.
- Always define the events explicitly before testing.
- Small sample spaces are often easiest to list completely.

Problem 3.7: Why independence does not depend on order

In higher studies of probability, statistics, and measure theory, independence is treated as a relationship between events rather than a directional process. So it should not depend on whether we name A first or B first.

Suppose event B is independent of event A , meaning that

$$P(B \cap A) = P(B)P(A).$$

Prove that event A is independent of event B .

Solution 3.7

Intersection is commutative, so

$$B \cap A = A \cap B.$$

Also, multiplication of real numbers is commutative, so

$$P(B)P(A) = P(A)P(B).$$

Therefore, from

$$P(B \cap A) = P(B)P(A),$$

we immediately get

$$P(A \cap B) = P(A)P(B).$$

This is exactly the condition for A to be independent of B .

Hence independence is symmetric.

Takeaways 3.7

- Independence is a relationship between events, not a directional action.
- The proof works because both intersection and multiplication are commutative.
- This symmetry is one reason independence extends naturally to random variables and more advanced probability structures.

Problem 3.8: Expected score of a game

A spinner gives scores according to the following distribution:

$$P(X = 0) = 0.2, \quad P(X = 2) = 0.5, \quad P(X = 5) = 0.3.$$

Find the expected score.

Solution 3.8

The expected value is

$$E(X) = \sum xP(X = x).$$

So

$$E(X) = 0(0.2) + 2(0.5) + 5(0.3) = 0 + 1 + 1.5 = 2.5.$$

Therefore the expected score is

$$\boxed{2.5}.$$

Takeaways 3.8

- Expected value is a weighted average.
- Multiply each outcome by its probability, then add.
- The expectation need not be an actual score on a single play.

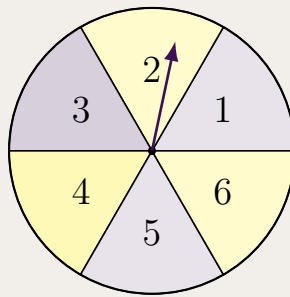
Problem 3.9: A fair n -sided spinner

A spinner has n equal sectors labelled

$$1, 2, 3, \dots, n.$$

Let X be the score obtained when the spinner is spun once.

Example. For $n = 6$, a fair spinner could look like this:



- Write down $P(X = k)$ for $1 \leq k \leq n$.
- Find $E(X)$.
- Find $\text{Var}(X)$.
- Interpret your result for $E(X)$ in terms of symmetry.

You may use

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}, \quad 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Solution 3.9

(a) Since the spinner is fair, each score is equally likely. Therefore

$$P(X = k) = \frac{1}{n} \quad \text{for } k = 1, 2, \dots, n.$$

(b) The expected value is

$$E(X) = \sum xP(X = x) = \sum_{k=1}^n k \cdot \frac{1}{n} = \frac{1}{n} \sum_{k=1}^n k.$$

Using

$$\sum_{k=1}^n k = \frac{n(n+1)}{2},$$

we get

$$E(X) = \frac{1}{n} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2}.$$

(c) First find

$$E(X^2) = \sum x^2 P(X = x) = \sum_{k=1}^n k^2 \cdot \frac{1}{n} = \frac{1}{n} \sum_{k=1}^n k^2.$$

Using

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6},$$

we have

$$E(X^2) = \frac{(n+1)(2n+1)}{6}.$$

So

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2.$$

Simplifying,

$$\text{Var}(X) = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{n^2 - 1}{12}.$$

(d) The value

$$E(X) = \frac{n+1}{2}$$

is exactly halfway between the smallest and largest possible scores, namely 1 and n . This matches the symmetry of a fair spinner: the distribution is balanced evenly around the centre.

Takeaways 3.9

- A fair finite model often leads to a discrete uniform distribution.
- Uniform distributions are natural examples where symmetry helps predict the mean.
- Variance can still be found systematically using $E(X^2) - [E(X)]^2$.

Problem 3.10: A random sum from a multiset

A student has six cards with values

$$4, 5, 6, 6, 7, 8.$$

Three cards are chosen at random without replacement, and their sum is recorded as the random variable X .

- (a) Construct the probability distribution table for X .
- (b) Find $E(X)$.
- (c) Without calculating $\text{Var}(X)$ fully, explain whether you would expect the spread of this distribution to be larger or smaller than if the six cards were

$$4, 5, 6, 7, 8, 9.$$

- (d) Comment briefly on why repeated values in the population affect the shape of the sampling distribution.

Solution 3.10

(a) Since 3 cards are chosen from 6, the number of equally likely selections is

$$\binom{6}{3} = 20.$$

Using the repeated value 6, we count how many selections produce each sum:

Sum	Selections
15	(4, 5, 6) in 2 ways
16	(4, 5, 7), (4, 6, 6)
17	(4, 5, 8), (4, 6, 7) in 2 ways, (5, 6, 6)
18	(4, 6, 8) in 2 ways, (5, 6, 7) in 2 ways
19	(4, 7, 8), (5, 6, 8) in 2 ways, (6, 6, 7)
20	(5, 7, 8), (6, 6, 8)
21	(6, 7, 8) in 2 ways

So the distribution table is

x	15	16	17	18	19	20	21
$P(X = x)$	$\frac{2}{20}$	$\frac{2}{20}$	$\frac{4}{20}$	$\frac{3}{20}$	$\frac{4}{20}$	$\frac{2}{20}$	$\frac{2}{20}$

This simplifies to

x	15	16	17	18	19	20	21
$P(X = x)$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{3}{20}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{10}$

(b) The expected value is

$$E(X) = 15 \left(\frac{1}{10} \right) + 16 \left(\frac{1}{10} \right) + 17 \left(\frac{1}{5} \right) + 18 \left(\frac{3}{20} \right) + 19 \left(\frac{1}{5} \right) + 20 \left(\frac{1}{10} \right) + 21 \left(\frac{1}{10} \right).$$

This gives

$$E(X) = 18.$$

(c) The spread should be **smaller** than for the cards

$$4, 5, 6, 7, 8, 9.$$

That is because the repeated value 6 creates more selections with central sums and fewer selections with widely spread values. Repetition pulls the distribution inward.

(d) Repeated values make some totals occur more often than others. This changes the shape of the sampling distribution by increasing clustering around certain sums. In higher studies of probability and statistics, this is an example of how the structure of a population affects the distribution of a statistic computed from a random sample.

Takeaways 3.10

- A random variable can be created by applying a rule, here “add the chosen cards”, to a combinatorial experiment.
- Equal-likelihood of selections does not mean equal-likelihood of the resulting sums.
- Repeated values in the underlying population can make the sampling distribution more concentrated around the centre.

Problem 3.11: Selection probability via combinations

From 6 girls and 4 boys, a committee of 3 students is chosen at random. Find the probability that the committee contains exactly 2 girls.

Solution 3.11

The total number of committees is

$$\binom{10}{3} = 120.$$

To have exactly 2 girls, choose 2 of the 6 girls and 1 of the 4 boys:

$$\binom{6}{2} \binom{4}{1} = 15 \cdot 4 = 60.$$

Therefore,

$$P(\text{exactly 2 girls}) = \frac{60}{120} = \frac{1}{2}.$$

Takeaways 3.11

- When order does not matter, combinations usually give the cleanest probability model.
- Count favourable selections and total selections in the same style.
- “Exactly” often suggests one precise combination pattern.

Problem 3.12: Leading digit of a height measurement

An anthropometric study measures the height of randomly selected adults in centimetres. Let H be the height of a randomly selected adult, and let X be the leading digit of H . For example, if

$$H = 175,$$

then

$$X = 1.$$

Assume all measured adults are between 100 cm and 250 cm tall.

- State whether X is a discrete random variable, a continuous random variable, or neither. Justify your answer.
- State the realistic sample space for X .
- A student argues that because almost every adult has a leading height digit of 1, the value of X is highly predictable and therefore X is not a valid random variable. Evaluate this claim.

Solution 3.12

(a) The variable X is a **discrete random variable**.

It is a random variable because its value is determined by a chance process: choosing an adult at random. It is discrete because it can take only separated whole-number values rather than a continuous range.

(b) Since all adults are between 100 cm and 250 cm tall, the leading digit can only be

1 or 2.

So the sample space for X is

$\{1, 2\}$.

(c) The student's claim is incorrect.

A random variable does *not* need to be uniformly distributed or hard to predict. It only needs to be determined by chance. Even if

$$P(X = 1)$$

is very large, there is still some chance that

$$X = 2$$

when a taller adult is selected. Since the exact outcome is not known with certainty before the selection is made, X remains a valid random variable.

Takeaways 3.12

- In mathematics, “random” means determined by chance, not necessarily evenly spread or highly unpredictable.
- A discrete random variable may still have a very skewed distribution.
- This problem also gives a useful contrast with Benford's Law: Benford behaviour often appears in data sets spread across several powers of ten, but adult human heights are trapped in a narrow range, so their leading digits are dominated by 1.

Problem 3.13: HSC 2023-style: Sample proportion and normal approximation

A recent census found that 30% of Australians were born overseas. A sample of 900 randomly selected Australians is surveyed.

Let $\hat{p}$ be the sample proportion of surveyed people who were born overseas.

(a) Show that

$$\text{Var}(\hat{p}) = \frac{7}{30000}.$$

(b) Use a normal approximation to estimate

$$P(\hat{p} \leq 0.31).$$

Solution 3.13

Let X be the number of surveyed people who were born overseas. Then

$$X \sim \text{Bin}(900, 0.3), \quad \hat{p} = \frac{X}{900}.$$

(a) Using the variance formula for a sample proportion,

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} = \frac{0.3(0.7)}{900} = \frac{0.21}{900} = \frac{21}{90000} = \frac{7}{30000}.$$

(b) Since n is large, we approximate

$$\hat{p} \approx N\left(0.3, \frac{7}{30000}\right).$$

So

$$\sigma_{\hat{p}} = \sqrt{\frac{7}{30000}} \approx 0.01528.$$

Now

$$P(\hat{p} \leq 0.31) \approx P\left(Z \leq \frac{0.31 - 0.30}{0.01528}\right) = P(Z \leq 0.654).$$

Hence

$$P(\hat{p} \leq 0.31) \approx 0.74.$$

So the required probability is approximately

$$\boxed{0.74}.$$

Takeaways 3.13

- Sample proportion is itself a random variable.
- Its mean is p and its variance is $\frac{p(1-p)}{n}$.
- For large samples, normal approximation turns proportion questions into standard normal questions.

3.3 Advanced Probability Problems**Problem 3.14: A probability table with unknown constant**

A discrete random variable X can take only the values 1, 2, 3, 4, with probability distribution

$$P(X = 1) = k, \quad P(X = 2) = 2k, \quad P(X = 3) = 3k, \quad P(X = 4) = 4k.$$

- Find k .
- Find $E(X)$.
- Find $\text{Var}(X)$.

Solution 3.14

(a) The probabilities must sum to 1, so

$$k + 2k + 3k + 4k = 10k = 1.$$

Hence

$$k = \frac{1}{10}.$$

(b) Then

$$P(X = 1) = \frac{1}{10}, \quad P(X = 2) = \frac{2}{10}, \quad P(X = 3) = \frac{3}{10}, \quad P(X = 4) = \frac{4}{10}.$$

Now

$$E(X) = 1 \cdot \frac{1}{10} + 2 \cdot \frac{2}{10} + 3 \cdot \frac{3}{10} + 4 \cdot \frac{4}{10} = \frac{1 + 4 + 9 + 16}{10} = 3.$$

(c) Next,

$$E(X^2) = 1^2 \cdot \frac{1}{10} + 2^2 \cdot \frac{2}{10} + 3^2 \cdot \frac{3}{10} + 4^2 \cdot \frac{4}{10} = \frac{1 + 8 + 27 + 64}{10} = 10.$$

Therefore,

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 10 - 3^2 = 1.$$

Takeaways 3.14

- A probability table must add to 1 before anything else.
- Variance is often quickest via $E(X^2) - [E(X)]^2$.
- Keep expectation and variance conceptually separate: centre versus spread.

Problem 3.15: Exactly two successes in a binomial setting

A biased coin lands heads with probability 0.7. The coin is tossed 5 times. Find the probability of getting exactly 2 heads.

Solution 3.15

Let X be the number of heads. Then

$$X \sim \text{Bin}(5, 0.7).$$

We want

$$P(X = 2) = \binom{5}{2} (0.7)^2 (0.3)^3.$$

Since

$$\binom{5}{2} = 10,$$

we obtain

$$P(X = 2) = 10(0.49)(0.027) = 0.1323.$$

Hence the probability is

$$\boxed{0.1323}.$$

Takeaways 3.15

- Check the four binomial conditions before applying the formula.
- The combination counts where the successes are placed.
- Keep the success probability attached to the event being counted.

Problem 3.16: At least one defective item

Each item produced by a machine is defective with probability 0.04, independently of other items. A sample of 8 items is chosen.

- Find the probability that exactly one item is defective.
- Find the probability that at least one item is defective.

Solution 3.16

Let X be the number of defective items. Then

$$X \sim \text{Bin}(8, 0.04).$$

- For exactly one defective item,

$$P(X = 1) = \binom{8}{1}(0.04)^1(0.96)^7.$$

So

$$P(X = 1) = 8(0.04)(0.96)^7 \approx 0.2409.$$

- For at least one defective item, use the complement:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.96)^8 \approx 0.2786.$$

Takeaways 3.16

- In binomial questions, “at least one” is usually a complement.
- Exact values and cumulative values often call for different tactics.
- Binomial modelling becomes natural when defect decisions are repeated independently.

Problem 3.17: A binomial distribution from a target score

In a multiple-choice quiz, each question has four options and exactly one correct answer. A student guesses every answer independently on 6 questions.

- Find the probability that the student answers exactly 4 questions correctly.
- Find the probability that the student answers at least 1 question correctly.

Solution 3.17

Let X be the number of correct answers. Since each question is guessed independently and

$$P(\text{correct}) = \frac{1}{4},$$

we have

$$X \sim \text{Bin}\left(6, \frac{1}{4}\right).$$

(a) Then

$$P(X = 4) = \binom{6}{4} \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^2.$$

Since

$$\binom{6}{4} = 15,$$

we get

$$P(X = 4) = 15 \cdot \frac{1}{256} \cdot \frac{9}{16} = \frac{135}{4096}.$$

(b) Using a complement,

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \left(\frac{3}{4}\right)^6.$$

Hence

$$P(X \geq 1) = 1 - \frac{729}{4096} = \frac{3367}{4096}.$$

Takeaways 3.17

- Guessing questions independently creates a natural binomial model.
- Exact counts use the binomial formula directly.
- “At least one” remains a complement even in a binomial setting.

Problem 3.18: Comparing two games by expectation

Game A pays \$8 with probability 0.2 and \$0 otherwise. Game B pays according to:

$$P(Y = 1) = 0.3, \quad P(Y = 4) = 0.4, \quad P(Y = 7) = 0.3.$$

Each game costs \$2 to play once. Determine which game has the higher expected profit.

Solution 3.18

Let A be the payout from Game A. Then

$$E(A) = 8(0.2) + 0(0.8) = 1.6.$$

So the expected profit from Game A is

$$1.6 - 2 = -0.4.$$

Now let Y be the payout from Game B. Then

$$E(Y) = 1(0.3) + 4(0.4) + 7(0.3) = 0.3 + 1.6 + 2.1 = 4.$$

So the expected profit from Game B is

$$4 - 2 = 2.$$

Therefore Game B has the higher expected profit.

Takeaways 3.18

- Separate payout from profit when a game has an entry cost.
- Expected value gives a fair comparison over many plays.
- A game can have an attractive top prize and still have poor expected profit.

Problem 3.19: HSC 2020-style: Biased coin and normal approximation

When a particular biased coin is tossed, the probability of obtaining a head is

$$\frac{3}{5}.$$

This coin is tossed 100 times. Let X be the number of heads obtained.

- Find $E(X)$.
- By finding $\text{Var}(X)$, show that the standard deviation of X is approximately 5.
- Use a normal approximation to estimate

$$P(55 \leq X \leq 65).$$

Solution 3.19

Since each toss is independent and the probability of a head is constant,

$$X \sim \text{Bin}\left(100, \frac{3}{5}\right).$$

(a) The expected value is

$$E(X) = np = 100 \cdot \frac{3}{5} = 60.$$

(b) The variance is

$$\text{Var}(X) = np(1-p) = 100 \cdot \frac{3}{5} \cdot \frac{2}{5} = 24.$$

Therefore

$$\sigma = \sqrt{24} \approx 4.90,$$

which is approximately 5.

(c) We use the normal approximation

$$X \approx N(60, 24).$$

Applying a continuity correction,

$$P(55 \leq X \leq 65) \approx P(54.5 \leq Y \leq 65.5),$$

where $Y \sim N(60, 24)$.

So

$$P(55 \leq X \leq 65) \approx P\left(\frac{54.5 - 60}{\sqrt{24}} \leq Z \leq \frac{65.5 - 60}{\sqrt{24}}\right).$$

Since

$$\frac{5.5}{\sqrt{24}} \approx 1.12,$$

we get

$$P(55 \leq X \leq 65) \approx P(-1.12 \leq Z \leq 1.12).$$

Using symmetry,

$$P(-1.12 \leq Z \leq 1.12) \approx 2(0.8686) - 1 = 0.7372.$$

Hence the required probability is approximately

$$\boxed{0.74}.$$

Takeaways 3.19

- Large binomial distributions are often approximated well by normal distributions.
- Mean and variance provide the centre and spread of the approximating normal model.
- Continuity correction improves the match between a discrete count and a continuous curve.

Problem 3.20: HSC 2022-style: When a normal approximation may fail

A manufacturer produces chocolate bars. Each bar has probability 0.3 of being underweight, independently of the others.

Sixteen bars are selected at random.

- (a) Let X be the number of underweight bars. State the distribution of X .
- (b) A student suggests using a normal approximation to estimate

$$P(X \geq 8).$$

Explain why this approximation may be unreliable.

Solution 3.20

- (a) Since there are 16 independent trials, each with two outcomes and constant success probability 0.3,

$$X \sim \text{Bin}(16, 0.3).$$

- (b) A normal approximation may be unreliable because the sample size is not especially large and the distribution is not close to symmetric. Here

$$np = 16(0.3) = 4.8, \quad n(1 - p) = 16(0.7) = 11.2.$$

The value of np is fairly small, so the binomial distribution is still noticeably skewed rather than well modelled by a symmetric normal curve.

Therefore the approximation to

$$P(X \geq 8)$$

may be poor, and an exact binomial calculation would be safer.

Takeaways 3.20

- Before using a normal approximation, check whether the binomial model is sufficiently well behaved.
- Small or skewed binomial distributions are not always approximated well by a normal curve.
- HSC questions may ask you to judge the model, not just calculate with it.

4 Part 2: Problems with Hints and Solutions (Concise)

Part 2 gives additional practice with shorter solutions. Each problem includes an upside-down hint so that you can choose how much support you want before checking the answer.

4.1 Basic Probability Problems

Problem 4.1: One card from a deck

A card is chosen at random from a standard deck. Find the probability that it is a king or a diamond.

Hint: Use the addition rule and subtract the overlap only once.

Solution 4.1

There are 4 kings and 13 diamonds, with one overlap: the king of diamonds.

$$P(\text{king or diamond}) = \frac{4 + 13 - 1}{52} = \frac{16}{52} = \frac{4}{13}.$$

Takeaways 4.1

- For a union, add the two probabilities and subtract the overlap once.
- The overlap here is the king of diamonds, which belongs to both events.

Problem 4.2: Three coin tosses

Three fair coins are tossed. Find the probability of getting exactly two heads.

Hint: List the successful patterns or use the binomial idea with $n = 3$ and $p = \frac{1}{2}$.

Solution 4.2

There are $\binom{3}{2} = 3$ ways to place two heads among three tosses.

$$P(\text{exactly two heads}) = \binom{3}{2} \left(\frac{1}{2}\right)^3 = \frac{3}{8}.$$

Takeaways 4.2

- In a repeated yes-no experiment, count the successful patterns before multiplying by the common probability.
- This is a small binomial-style question with $n = 3$ and $p = \frac{1}{2}$.

Problem 4.3: Without replacement

A box contains 5 white counters and 2 black counters. Two counters are drawn without replacement. Find the probability that both are white.

Hint: Multiply the changing fractions directly.

Solution 4.3

$$P(\text{both white}) = \frac{5}{7} \cdot \frac{4}{6} = \frac{20}{42} = \frac{10}{21}.$$

Takeaways 4.3

- Without replacement, the second probability changes because the box contents change.
- Multiplying stage-by-stage is often the quickest method for short draws.

Problem 4.4: Expected value from a table

A random variable X has

$$P(X = 1) = 0.4, \quad P(X = 2) = 0.1, \quad P(X = 4) = 0.5.$$

Find $E(X)$.

Hint: Multiply each outcome by its probability and add.

Solution 4.4

$$E(X) = 1(0.4) + 2(0.1) + 4(0.5) = 0.4 + 0.2 + 2 = 2.6.$$

Takeaways 4.4

- Expected value is a weighted average, not just an ordinary mean of the outcomes.
- Multiply each value by its probability, then add.

Problem 4.5: Hannah and the snack boxes

Hannah may choose one item from a snack shop. The shop has:

- 6 boxes of muffins, each muffin costing \$3.20
- 5 boxes of cookies, each cookie costing \$2.40
- 4 boxes of juice boxes, each juice box costing \$2.80

Hannah chooses a box at random, then takes one item from that box. Find the expected cost of the item she takes.

Hint: First write down the probability of choosing each type of box. Then use a weighted average.

Solution 4.5

There are

$$6 + 5 + 4 = 15$$

boxes in total.

So

$$P(\text{muffin}) = \frac{6}{15}, \quad P(\text{cookie}) = \frac{5}{15}, \quad P(\text{juice box}) = \frac{4}{15}.$$

Let X be the cost of the item chosen. Then

$$E(X) = 3.20 \left(\frac{6}{15} \right) + 2.40 \left(\frac{5}{15} \right) + 2.80 \left(\frac{4}{15} \right).$$

$$E(X) = 1.28 + 0.80 + \frac{11.2}{15} = 1.28 + 0.80 + 0.746\bar{6} = 2.826\bar{6}.$$

Therefore the expected cost is

\$2.83 (approximately).

Takeaways 4.5

- Turn the box counts into probabilities before finding the expected cost.
- In modelling questions, keep the distinction clear between an item's cost and its selection probability.

Problem 4.6: Reading probabilities from a table

A random variable X has the following probability distribution:

x	0	1	2	3	4	5
$P(X = x)$	0.1	0.2	0.3	0.25	0.05	0.1

Find:

- $P(X \text{ is odd}),$
- $P(X \geq 2).$

Hint: Add the probabilities of all outcomes that satisfy each condition.

Solution 4.6

(i) The odd values are 1, 3, and 5, so

$$P(X \text{ is odd}) = P(X = 1) + P(X = 3) + P(X = 5) = 0.2 + 0.25 + 0.1 = 0.55.$$

(ii) The values at least 2 are 2, 3, 4, and 5, so

$$P(X \geq 2) = P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) = 0.3 + 0.25 + 0.05 + 0.1 = 0.7.$$

Takeaways 4.6

- When a distribution table is given, answer event questions by adding the probabilities of all matching outcomes.
- Read the condition carefully: “odd” and “at least 2” select different sets of x -values.

Problem 4.7: Complement of an impossible pair

Two fair dice are rolled. Find the probability that the sum is not 7.

Hint: There are six outcomes with sum 7.

Solution 4.7

So

$$P(\text{sum } 7) = \frac{6}{36} = \frac{1}{6}.$$

$$P(\text{sum not } 7) = 1 - \frac{1}{6} = \frac{5}{6}.$$

Takeaways 4.7

- “Not” often signals a complement.
- If the complement is easier to count, find it first and subtract from 1.

Problem 4.8: Single-stage conditional probability

In a box there are 3 green counters and 5 yellow counters. One counter is drawn. Given that it is not yellow, find the probability that it is green.

Hint: If it is not yellow, what possibilities remain?

Solution 4.8

If the counter is not yellow, it must be green. Therefore

$$P(\text{green} \mid \text{not yellow}) = 1.$$

Takeaways 4.8

- In conditional probability, first reduce the sample space to the given condition.
- If only one outcome remains possible, the conditional probability is 1.

Problem 4.9: Binomial single success

A fair coin is tossed 5 times. Find the probability of exactly one head.

Hint: Use $\binom{5}{1} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^4$.

Solution 4.9

$$P(X = 1) = \binom{5}{1} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^4 = 5 \cdot \frac{1}{32} = \frac{5}{32}.$$

Takeaways 4.9

- “Exactly one” in repeated trials usually leads to a binomial coefficient.
- The coefficient counts where the single success can occur.

Problem 4.10: One coin and one die

A fair coin and a fair die are used. Find the probability of getting a head and an even number.

Hint: These two devices act independently, so multiply the probabilities.

Solution 4.10

$$P(\text{head and even}) = \frac{1}{2} \cdot \frac{3}{6} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Takeaways 4.10

- For independent devices, multiply the separate probabilities.
- A coin-and-die question is a classic independence model.

Problem 4.11: No red counter

A bag contains 3 red counters and 4 blue counters. One counter is drawn. Find the probability that it is not red.

Hint: Either count blue directly or use the complement of red.

Solution 4.11

$$P(\text{not red}) = 1 - \frac{3}{7} = \frac{4}{7}.$$

Takeaways 4.11

- Complements are especially efficient in one-draw questions.
- Here “not red” is the same event as drawing blue.

Problem 4.12: A letter from MIRANDA

A letter is chosen at random from the word MIRANDA. Find the probability that it is:

- an A,
- a vowel,
- a letter of the word HACKING.

Hint: There are 7 letters in total. Count favourable letters carefully, including repeated letters when they appear in the word.

Solution 4.12

The word MIRANDA has 7 letters:

M, I, R, A, N, D, A.

- (i) There are 2 A's, so

$$P(A) = \frac{2}{7}.$$

- (ii) The vowels are I, A, A, so there are 3 vowels. Thus

$$P(\text{vowel}) = \frac{3}{7}.$$

- (iii) The letters in HACKING are

H, A, C, K, I, N, G.

Among the letters of MIRANDA, the ones that belong to HACKING are

I, A, N, A,

so there are 4 favourable letters. Hence

$$P(\text{a letter of HACKING}) = \frac{4}{7}.$$

Takeaways 4.12

- In equally likely letter questions, each position in the word counts, including repeated letters.
- Be careful not to switch from counting letters to counting distinct symbols.

Problem 4.13: HSC 2025-style: Bernoulli mean and variance

A Bernoulli random variable X has

$$P(X = 1) = \frac{2}{3}, \quad P(X = 0) = \frac{1}{3}.$$

Find $E(X)$ and $\text{Var}(X)$.

Hint: For a Bernoulli random variable with success probability p , use $E(X) = p$ and $\text{Var}(X) = p(1-p)$.

Solution 4.13

Here $p = \frac{2}{3}$, so

$$E(X) = \frac{2}{3}, \quad \text{Var}(X) = \frac{2}{3} \cdot \frac{1}{3} = \frac{2}{9}.$$

Takeaways 4.13

- A Bernoulli random variable takes only the values 0 and 1.
- Its mean and variance are controlled entirely by the success probability p .

4.2 Medium Probability Problems

Problem 4.14: Conditional selection

From a standard deck, one card is selected. Given that the card is red, find the probability that it is a queen.

Hint: Restrict the sample space to the red cards first.

Solution 4.14

There are 26 red cards and 2 red queens.

$$P(\text{queen} \mid \text{red}) = \frac{2}{26} = \frac{1}{13}.$$

Takeaways 4.14

- Conditional probability starts by shrinking the sample space to the given condition.
- After conditioning on “red”, only the red cards matter.

Problem 4.15: Independence check

A fair die is rolled once. Let

$$A = \{\text{an even number}\}, \quad B = \{\text{a number greater than 3}\}.$$

Determine whether A and B are independent.

Hint: Compute $P(A)$, $P(B)$, and $P(A \cap B)$, then compare.

Solution 4.15

$$P(A) = \frac{3}{6} = \frac{1}{2}, \quad P(B) = \frac{3}{6} = \frac{1}{2}.$$

Also,

$$A \cap B = \{4, 6\},$$

so

$$P(A \cap B) = \frac{2}{6} = \frac{1}{3}.$$

Since

$$\frac{1}{3} \neq \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4},$$

the events are not independent.

Takeaways 4.15

- To test independence, compare $P(A \cap B)$ with $P(A)P(B)$.
- Similar-looking events are not automatically independent.

Problem 4.16: Finding a missing probability

A random variable X has

$$P(X = 0) = 0.2, \quad P(X = 1) = 0.35, \quad P(X = 2) = p, \quad P(X = 3) = 0.15.$$

Find p and then find $E(X)$.

Hint: Probabilities must add to 1 before you calculate expectation.

Solution 4.16

$$0.2 + 0.35 + p + 0.15 = 1$$

gives

$$p = 0.30.$$

Then

$$E(X) = 0(0.2) + 1(0.35) + 2(0.30) + 3(0.15) = 0.35 + 0.60 + 0.45 = 1.40.$$

Takeaways 4.16

- A probability distribution must add to 1 before any expectation calculation is valid.
- Solve for missing probabilities first, then use the weighted-average formula.

Problem 4.17: Binomial mean and variance

If

$$X \sim \text{Bin}(12, 0.25),$$

find $E(X)$ and $\text{Var}(X)$.

Hint: Use the standard formulas for a binomial random variable.

Solution 4.17

$$E(X) = np = 12(0.25) = 3.$$

$$\text{Var}(X) = np(1 - p) = 12(0.25)(0.75) = 2.25.$$

Takeaways 4.17

- Once a binomial model is identified, mean and variance come straight from the standard formulas.
- Always check that both n and p have been read correctly from the question.

Problem 4.18: A union from a survey table

In a group of 40 students, 24 study French, 18 study Music, and 10 study both. One student is selected at random. Find the probability that the student studies French or Music.

Hint: Use the addition rule with counts.

Solution 4.18

$$P(F \cup M) = \frac{24 + 18 - 10}{40} = \frac{32}{40} = \frac{4}{5}.$$

Takeaways 4.18

- For a union, add the two groups and subtract the overlap once.
- Counts can be handled first, then converted to probability at the end.

Problem 4.19: Variance from a probability table

A random variable X has

$$P(X = 0) = 0.5, \quad P(X = 2) = 0.3, \quad P(X = 4) = 0.2.$$

Find $\text{Var}(X)$.

Hint: First find $E(X)$ and $E(X^2)$.

Solution 4.19

$$E(X) = 0(0.5) + 2(0.3) + 4(0.2) = 1.4.$$

$$E(X^2) = 0^2(0.5) + 2^2(0.3) + 4^2(0.2) = 4.4.$$

So

$$\text{Var}(X) = 4.4 - (1.4)^2 = 4.4 - 1.96 = 2.44.$$

Takeaways 4.19

- Variance from a table usually needs both $E(X)$ and $E(X^2)$.
- Keep the square on the value of X , not on the probability.

Problem 4.20: Distribution table from draws without replacement

A bag contains six balls marked

$$0, 1, 2, 3, 4, 5.$$

Three balls are drawn in succession, without replacement, and the number of odd-numbered balls drawn is recorded. Let X be the number of odd-numbered balls chosen. Draw up a probability distribution table for X .

$$\frac{\binom{3}{3} \binom{3}{0}}{\binom{6}{3}} = P(X=3)$$

Hint: There are 3 odd-numbered balls and 3 even-numbered balls. Use combinations.

Solution 4.20

The odd-numbered balls are

$$1, 3, 5,$$

and the even-numbered balls are

$$0, 2, 4.$$

Since 3 balls are drawn, the possible values of X are

$$0, 1, 2, 3.$$

There are

$$\binom{6}{3} = 20$$

equally likely selections of 3 balls.

When $X = 0$: all 3 chosen balls are even, so

$$P(X = 0) = \frac{\binom{3}{0}\binom{3}{3}}{\binom{6}{3}} = \frac{1}{20}.$$

When $X = 1$: choose 1 odd and 2 even, so

$$P(X = 1) = \frac{\binom{3}{1}\binom{3}{2}}{\binom{6}{3}} = \frac{9}{20}.$$

When $X = 2$: choose 2 odd and 1 even, so

$$P(X = 2) = \frac{\binom{3}{2}\binom{3}{1}}{\binom{6}{3}} = \frac{9}{20}.$$

When $X = 3$: all 3 chosen balls are odd, so

$$P(X = 3) = \frac{\binom{3}{3}\binom{3}{0}}{\binom{6}{3}} = \frac{1}{20}.$$

Therefore the probability distribution table is

x	0	1	2	3
$P(X = x)$	$\frac{1}{20}$	$\frac{9}{20}$	$\frac{9}{20}$	$\frac{1}{20}$

Takeaways 4.20

- This is a without-replacement counting problem, so combinations are the natural tool.
- A full distribution table should list all possible values of X and probabilities that add to 1.

Problem 4.21: A first binomial distribution table

A fair coin is tossed 4 times and the number of heads is recorded. Construct a table showing the probability distribution of X , the number of heads, and calculate $E(X)$.

Note. This is an example of a binomial distribution: a multi-stage experiment with identical stages, where at each stage there are two possible outcomes. Here there are 4 stages because the coin is tossed 4 times, and each stage has two possible outcomes, heads or tails.

$$\binom{4}{r} \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{4-r} = P(X=r)$$

Use

$$X \sim \text{Bin}\left(4, \frac{1}{2}\right)$$

Hint: For a fair coin,**Solution 4.21**

Since the coin is fair and tossed 4 times independently,

$$X \sim \text{Bin}\left(4, \frac{1}{2}\right).$$

So for $r = 0, 1, 2, 3, 4$,

$$P(X=r) = \binom{4}{r} \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{4-r} = \binom{4}{r} \left(\frac{1}{2}\right)^4 = \frac{\binom{4}{r}}{16}.$$

Therefore

x	0	1	2	3	4
$P(X=x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

Now

$$E(X) = 0 \left(\frac{1}{16}\right) + 1 \left(\frac{4}{16}\right) + 2 \left(\frac{6}{16}\right) + 3 \left(\frac{4}{16}\right) + 4 \left(\frac{1}{16}\right).$$

So

$$E(X) = \frac{0 + 4 + 12 + 12 + 4}{16} = \frac{32}{16} = 2.$$

Hence the expected number of heads is

$$\boxed{2}.$$

Takeaways 4.21

- A binomial distribution table comes from choosing where the heads occur.
- The expected value matches the shortcut formula $E(X) = np$.

Problem 4.22: A first hypergeometric distribution table

Two cards are drawn in succession from a standard deck, without replacement, and the number of hearts is recorded. Construct a table showing the probability distribution of X , the number of hearts drawn, and calculate $E(X)$.

Note. This is an example of a hypergeometric distribution: a multi-stage experiment with two possible outcomes at each stage, where selection is made without replacement. Here there are 2 stages because two cards are drawn.

Hint: The possible values are 0, 1, and 2. Use combinations, or count the three cases directly.

Solution 4.22

There are 13 hearts and 39 non-hearts in a standard deck of 52 cards.
The possible values of X are

$$0, 1, 2.$$

Using combinations,

$$P(X = 0) = \frac{\binom{39}{2}}{\binom{52}{2}} = \frac{741}{1326} = \frac{247}{442},$$

$$P(X = 1) = \frac{\binom{13}{1}\binom{39}{1}}{\binom{52}{2}} = \frac{507}{1326} = \frac{169}{442},$$

and

$$P(X = 2) = \frac{\binom{13}{2}}{\binom{52}{2}} = \frac{78}{1326} = \frac{13}{221}.$$

Therefore the probability distribution table is

x	0	1	2
$P(X = x)$	$\frac{247}{442}$	$\frac{169}{442}$	$\frac{13}{221}$

Now

$$E(X) = 0 \left(\frac{247}{442} \right) + 1 \left(\frac{169}{442} \right) + 2 \left(\frac{13}{221} \right).$$

So

$$E(X) = \frac{169}{442} + \frac{26}{221} = \frac{169}{442} + \frac{52}{442} = \frac{221}{442} = \frac{1}{2}.$$

Hence the expected number of hearts is

$$\boxed{\frac{1}{2}}.$$

Takeaways 4.22

- Hypergeometric questions model sampling without replacement.
- This contrasts with binomial models, where trials are independent and replacement is implicit.

Problem 4.23: At least two heads

A biased coin lands heads with probability 0.4. It is tossed 4 times. Find the probability of at least two heads.

Hint: Add $P(X = 2)$, $P(X = 3)$, and $P(X = 4)$, or subtract $P(X = 0)$ and $P(X = 1)$.

Solution 4.23

Let $X \sim \text{Bin}(4, 0.4)$. Then

$$P(X \geq 2) = 1 - [P(X = 0) + P(X = 1)].$$

Now

$$P(X = 0) = (0.6)^4 = 0.1296,$$

and

$$P(X = 1) = \binom{4}{1}(0.4)(0.6)^3 = 4(0.4)(0.216) = 0.3456.$$

Hence

$$P(X \geq 2) = 1 - 0.1296 - 0.3456 = 0.5248.$$

Takeaways 4.23

- “At least” questions are often quickest by using the complement.
- For binomial probabilities, check whether adding a few terms or subtracting a few terms is shorter.

Problem 4.24: Exactly one black card

Two cards are drawn from a standard deck without replacement. Find the probability that exactly one card is black.

Hint: Count one black and one red in either order, or use combinations.

Solution 4.24

Using combinations,

$$P(\text{exactly one black}) = \frac{\binom{26}{1}\binom{26}{1}}{\binom{52}{2}} = \frac{676}{1326} = \frac{26}{51}.$$

Takeaways 4.24

- “Exactly one of each type” is a strong sign to use combinations.
- Without replacement, the order does not matter when you count selections with combinations.

Problem 4.25: Expected number of tails

A fair coin is tossed 9 times. Let X be the number of tails. Find $E(X)$.

Hint: Recognise a binomial model.

Solution 4.25

so

$$X \sim \text{Bin}\left(9, \frac{1}{2}\right),$$

$$E(X) = np = 9 \cdot \frac{1}{2} = \frac{9}{2} = 4.5.$$

Takeaways 4.25

- For binomial counts, the mean is the number of trials times the success probability.
- The expected number of tails need not be an integer outcome.

Problem 4.26: Fair coin tossed n times

If a fair coin is tossed n times, where $n > 1$, find the probability of obtaining:

- n tails,
- at least two tails,
- no more than half of the tosses as tails.

Hint: Let X be the number of tails, so $X \sim \text{Bin}\left(n, \frac{1}{2}\right)$. For part (iii), use symmetry about the middle and treat odd and even n separately.

Solution 4.26

Let

$$X \sim \text{Bin}\left(n, \frac{1}{2}\right),$$

where X is the number of tails.

(i) Getting n tails means every toss is a tail, so

$$P(X = n) = \left(\frac{1}{2}\right)^n.$$

(ii) “At least two tails” means $X \geq 2$, so

$$P(X \geq 2) = 1 - [P(X = 0) + P(X = 1)].$$

Now

$$P(X = 0) = \left(\frac{1}{2}\right)^n$$

and

$$P(X = 1) = \binom{n}{1} \left(\frac{1}{2}\right)^n = \frac{n}{2^n}.$$

Hence

$$P(X \geq 2) = 1 - \frac{1+n}{2^n}.$$

(iii) We want

$$P\left(X \leq \frac{n}{2}\right).$$

Because the coin is fair, the binomial distribution is symmetric.

If n is odd, there is no middle value, so exactly half the outcomes lie on each side:

$$P\left(X \leq \frac{n}{2}\right) = \frac{1}{2}.$$

If n is even, the middle value $X = \frac{n}{2}$ must be shared equally between the two sides, so

$$P\left(X \leq \frac{n}{2}\right) = \frac{1}{2} + \frac{1}{2}P\left(X = \frac{n}{2}\right).$$

Therefore

$$P\left(X \leq \frac{n}{2}\right) = \frac{1}{2} + \frac{1}{2}\binom{n}{n/2} \left(\frac{1}{2}\right)^n = \frac{1}{2} + \frac{1}{2^{n+1}}\binom{n}{n/2}.$$

Takeaways 4.26

- Symmetry is a powerful shortcut for $\text{Bin}(n, \frac{1}{2})$.
- Odd and even n behave differently because only even n has a central term at $X = \frac{n}{2}$.

Problem 4.27: HSC 2024-style: Driver test

To pass a driver knowledge test, a learner must answer at least 9 out of 10 questions correctly.

A learner already knows the answers to 5 questions and guesses randomly on the remaining 5 questions. Each guessed question has 4 options, with exactly one correct answer.

Find the probability that the learner passes the test.

Hint: The learner already has 5 marks, so passing means getting at least 4 of the 5 guessed questions correct.

Solution 4.27

Let X be the number of guessed questions answered correctly. Then

$$X \sim \text{Bin}\left(5, \frac{1}{4}\right).$$

To pass, the learner needs at least 4 correct guesses:

$$P(\text{pass}) = P(X \geq 4) = P(X = 4) + P(X = 5).$$

$$P(X = 4) = \binom{5}{4} \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right) = 5 \cdot \frac{1}{256} \cdot \frac{3}{4} = \frac{15}{1024},$$

and

$$P(X = 5) = \left(\frac{1}{4}\right)^5 = \frac{1}{1024}.$$

So

$$P(\text{pass}) = \frac{15}{1024} + \frac{1}{1024} = \frac{16}{1024} = \frac{1}{64}.$$

Takeaways 4.27

- Translate the real situation into a binomial count before calculating.
- Here the known answers shift the target: passing means at least 4 correct guesses, not 9 guessed questions.

Problem 4.28: HSC 2024-style: Same-metal coins

A bag contains 7 silver coins and 3 bronze coins. Two coins are drawn without replacement. Find the probability that both coins are made of the same metal.

Hint: Add the probabilities of both silver and both bronze.

Solution 4.28

$$P(\text{same metal}) = P(\text{both silver}) + P(\text{both bronze}).$$

$$P(\text{both silver}) = \frac{7}{10} \cdot \frac{6}{9} = \frac{42}{90}, \quad P(\text{both bronze}) = \frac{3}{10} \cdot \frac{2}{9} = \frac{6}{90}.$$

Therefore

$$P(\text{same metal}) = \frac{42}{90} + \frac{6}{90} = \frac{48}{90} = \frac{8}{15}.$$

Takeaways 4.28

- “Same metal” splits naturally into two disjoint cases: both silver or both bronze.
- When cases cannot happen together, their probabilities can be added directly.

Problem 4.29: HSC 2023-style: Mixed equipment probabilities

A gym has 5 treadmills and 4 rowing machines. On average, each treadmill is in use with probability 0.65 and each rowing machine is in use with probability 0.40, independently of the others.

- Find an expression for the probability that exactly 3 treadmills are in use.
- Find an expression for the probability that exactly 3 treadmills are in use and no rowing machines are in use.

Hint: Use a binomial expression for the treadmills, then multiply by the rowing-machine condition.

Solution 4.29

- (a) For the 5 treadmills,

$$P(\text{exactly 3 in use}) = \binom{5}{3} (0.65)^3 (0.35)^2.$$

- (b) The probability that no rowing machines are in use is

$$(0.60)^4.$$

So the required expression is

$$\binom{5}{3} (0.65)^3 (0.35)^2 (0.60)^4.$$

Takeaways 4.29

- Mixed-device questions may need more than one model in the same expression.
- Independence lets you multiply the treadmill event by the rowing-machine event.

Problem 4.30: HSC 2021-style: A Bernoulli complement

The random variable X represents the number of successes in 10 independent Bernoulli trials, where the probability of success in each trial is 0.9.

If

$$r = P(X \geq 1),$$

determine whether r is greater than, equal to, or less than 0.9.

Hint: Use the complement $P(X \geq 1) = 1 - P(X = 0)$.

Solution 4.30

$$r = P(X \geq 1) = 1 - P(X = 0) = 1 - (0.1)^{10}.$$

Since $(0.1)^{10}$ is extremely small,

$$r = 1 - (0.1)^{10} > 0.9.$$

Takeaways 4.30

- “At least one” is often easiest through the complement “none”.
- A large success probability over many trials pushes $P(X \geq 1)$ very close to 1.

4.3 Advanced Probability Problems**Problem 4.31: No sixes in repeated rolls**

A fair die is rolled 6 times. Find the probability that no six appears.

Hint: This is repeated independent failure with probability $\frac{5}{6}$ each time.

Solution 4.31

$$P(\text{no sixes}) = \left(\frac{5}{6}\right)^6 = \frac{15625}{46656}.$$

Takeaways 4.31

- Repeated independent failures multiply.
- “No sixes” means using the failure probability $\frac{5}{6}$ on every roll.

Problem 4.32: At most two successes

If

$$X \sim \text{Bin}(4, 0.3),$$

find $P(X \leq 2)$.

Hint: Either add $P(X = 0)$, $P(X = 1)$, and $P(X = 2)$, or subtract the complement.

Solution 4.32

$$P(X \leq 2) = 1 - [P(X = 3) + P(X = 4)].$$

Now

$$P(X = 3) = \binom{4}{3} (0.3)^3 (0.7) = 4(0.027)(0.7) = 0.0756,$$

and

$$P(X = 4) = (0.3)^4 = 0.0081.$$

Hence

$$P(X \leq 2) = 1 - 0.0756 - 0.0081 = 0.9163.$$

Takeaways 4.32

- For tail probabilities, complements often reduce the amount of arithmetic.
- In binomial questions, choose the shorter side of the distribution to sum.

Problem 4.33: Game value

A game costs \$3 to play. A fair die is rolled. If a 6 appears, the player receives \$10. Otherwise the player receives nothing. Find the expected profit for the player.

Hint: Define profit, not payout.

Solution 4.33

Let X be the player's profit.

If a 6 appears, profit is

$$10 - 3 = 7$$

with probability $\frac{1}{6}$.

Otherwise profit is

$$-3$$

with probability $\frac{5}{6}$.

So

$$E(X) = 7 \left(\frac{1}{6} \right) + (-3) \left(\frac{5}{6} \right) = \frac{7}{6} - \frac{15}{6} = -\frac{8}{6} = -\frac{4}{3}.$$

Thus the expected profit is

$$\boxed{-\frac{4}{3} \text{ dollars}}.$$

Takeaways 4.33

- Expected profit is not the same as expected payout.
- Always define the random variable carefully before finding its mean.

Problem 4.34: Variance from a small distribution

A discrete random variable X has

$$P(X = 1) = 0.2, \quad P(X = 3) = 0.5, \quad P(X = 5) = 0.3.$$

Find $\text{Var}(X)$.

Hint: Compute $E(X)$ and $E(X^2)$ separately.

Solution 4.34

$$E(X) = 1(0.2) + 3(0.5) + 5(0.3) = 0.2 + 1.5 + 1.5 = 3.2.$$

$$E(X^2) = 1^2(0.2) + 3^2(0.5) + 5^2(0.3) = 0.2 + 4.5 + 7.5 = 12.2.$$

Therefore,

$$\text{Var}(X) = 12.2 - (3.2)^2 = 12.2 - 10.24 = 1.96.$$

Takeaways 4.34

- Variance measures spread, so it is built from both the first and second moments.
- A clean table method is $E(X^2) - [E(X)]^2$.

Problem 4.35: Exactly three correct guesses

A student guesses on 7 true-false questions independently. Find the probability of getting exactly 3 correct.

Hint: This is binomial with $p = \frac{1}{2}$.

Solution 4.35

$$P(X = 3) = \binom{7}{3} \left(\frac{1}{2}\right)^7 = 35 \cdot \frac{1}{128} = \frac{35}{128}.$$

Takeaways 4.35

- A true-false guessing model is binomial with $p = \frac{1}{2}$.
- The combination counts where the 3 correct answers occur.

Problem 4.36: Expected number of red balls

A box contains 5 red balls and 7 white balls. Three balls are drawn with replacement. Let X be the number of red balls drawn. Find $E(X)$ and $\text{Var}(X)$.

Hint: With replacement, the number of reds is binomial.

Solution 4.36

Here

$$X \sim \text{Bin}\left(3, \frac{5}{12}\right).$$

Therefore

$$E(X) = np = 3 \cdot \frac{5}{12} = \frac{5}{4},$$

and

$$\text{Var}(X) = np(1-p) = 3 \cdot \frac{5}{12} \cdot \frac{7}{12} = \frac{35}{48}.$$

Takeaways 4.36

- With replacement, the trials are independent, so a binomial model is appropriate.
- Mean and variance both come from the same success probability $p = \frac{5}{12}$.

Problem 4.37: A fair game check

A game pays \$12 with probability 0.15, \$5 with probability 0.25, and \$0 otherwise. What entry fee would make the game fair?

Hint: A fair game has expected profit 0, so the fee must equal expected payout.

Solution 4.37

The expected payout is

$$12(0.15) + 5(0.25) + 0(0.60) = 1.8 + 1.25 = 3.05.$$

So the fair entry fee is

$$\boxed{\$3.05}.$$

Takeaways 4.37

- A fair game has expected profit 0, so the fee must equal the expected payout.
- Fair in the mathematical sense does not mean players always break even in practice.

Problem 4.38: More than half successes

If

$$X \sim \text{Bin}(5, 0.2),$$

find $P(X > 2)$.

Hint: Add $P(X = 3)$, $P(X = 4)$, and $P(X = 5)$.

Solution 4.38

$$P(X > 2) = P(X = 3) + P(X = 4) + P(X = 5).$$

Now

$$P(X = 3) = \binom{5}{3}(0.2)^3(0.8)^2 = 10(0.008)(0.64) = 0.0512,$$

$$P(X = 4) = \binom{5}{4}(0.2)^4(0.8) = 5(0.0016)(0.8) = 0.0064,$$

$$P(X = 5) = (0.2)^5 = 0.00032.$$

Therefore

$$P(X > 2) = 0.0512 + 0.0064 + 0.00032 = 0.05792.$$

Takeaways 4.38

- “More than half” becomes a right-tail binomial probability.
- Writing the relevant values first helps avoid missing a term.

Problem 4.39: Conditional probability after a draw

A box contains 4 defective bulbs and 9 good bulbs. Two bulbs are drawn without replacement. Given that the first bulb drawn is defective, find the probability that the second bulb is good.

Hint: Condition on the first draw and update the box contents.

Solution 4.39

If the first bulb is defective, then 3 defective and 9 good bulbs remain, so 12 bulbs remain in total. Hence

$$P(\text{second good} \mid \text{first defective}) = \frac{9}{12} = \frac{3}{4}.$$

Takeaways 4.39

- Once a condition is given, update the contents before calculating.
- Conditional probability after a draw is often just a revised fraction.

Problem 4.40: Geometric probability in a rectangular field

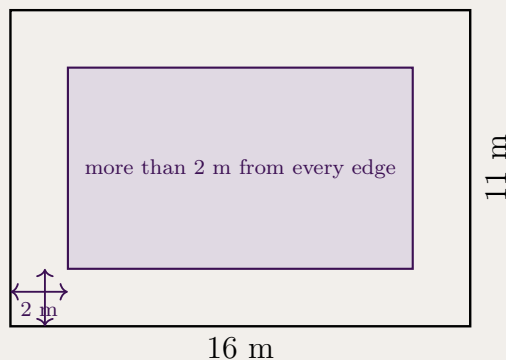
An $11\text{ m} \times 16\text{ m}$ rectangular field is considered, and a sheep wanders randomly around the field.

Find the probability that the sheep is:

- (a) more than 2 m from the edge of the field,

Note. This is a geometric probability question. If every point in the field is equally likely, then

$$P(\text{event}) = \frac{\text{favourable area}}{\text{total area}}.$$



Hint: For part (a), imagine shrinking the rectangle by 2 m from every edge.

Solution 4.40

The total area of the field is

$$11 \times 16 = 176\text{ m}^2.$$

(a) Being more than 2 m from the edge means being inside the inner rectangle left after removing a 2 m strip from each side.

Its dimensions would be

$$11 - 4 = 7 \quad \text{and} \quad 16 - 4 = 12,$$

so its area is

$$7 \times 12 = 84\text{ m}^2.$$

Therefore,

$$P(\text{more than 2 m from the edge}) = \frac{84}{176} = \frac{21}{44}.$$

Takeaways 4.40

- Geometric probability replaces counting by comparing favourable area with total area.
- When a point must stay a fixed distance from every edge, the favourable region is an inner rectangle.

Problem 4.41: Coin on a square grid

A coin with radius 1 cm is thrown down on a square grid whose lines are 6 cm apart. Find the probability that the coin does not intersect any grid line.

Note. This is a geometric probability question. By symmetry, it is enough to study the position of the centre of the coin inside one $6\text{ cm} \times 6\text{ cm}$ square.

Hint: The coin avoids all grid lines exactly when its centre lands more than 1 cm away from every side of the square.

Solution 4.41

Consider one square cell of the grid, with side length 6 cm. The centre of the coin is equally likely to land anywhere in this square.

For the coin not to touch a grid line, its centre must be at least 1 cm from each side. So the centre must lie inside the inner square obtained by removing a strip of width 1 cm from each edge.

That inner square has side length

$$6 - 2(1) = 4,$$

so its area is

$$4^2 = 16.$$

The total area of the square cell is

$$6^2 = 36.$$

Therefore

$$P(\text{coin does not intersect a grid line}) = \frac{16}{36} = \frac{4}{9}.$$

Takeaways 4.41

- In grid problems, track the position of the centre rather than the whole object.
- The probability comes from the area of the safe inner region inside one cell.

Problem 4.42: HSC 2024-style: Sample size from standard deviation

Suppose $\hat{p}$ is the sample proportion of successes from n independent Bernoulli trials with success probability $p = 0.5$.

Find the smallest value of n such that the standard deviation of $\hat{p}$ is less than 0.06.

$$\frac{u}{(d-1)d} = (d)^{\text{re}\Lambda}$$

Hint: Use

Solution 4.42

Since $p = 0.5$,

$$\text{Var}(\hat{p}) = \frac{0.5(0.5)}{n} = \frac{0.25}{n}.$$

So

$$\sigma_{\hat{p}} = \sqrt{\frac{0.25}{n}} = \frac{0.5}{\sqrt{n}}.$$

We require

$$\frac{0.5}{\sqrt{n}} < 0.06.$$

Hence

$$\sqrt{n} > \frac{0.5}{0.06} = \frac{25}{3},$$

so

$$n > \frac{625}{9} \approx 69.44.$$

Therefore the smallest integer value is

$$\boxed{70}.$$

Takeaways 4.42

- Sample proportion questions often reduce to controlling the standard deviation.
- After solving the inequality, round up because the sample size must be an integer.

Problem 4.43: HSC 2024-style: Charity donations

A charity worker talks to exactly 100 people in a day. The probability that any given person makes a donation is 0.31, independently of the others.

Use a normal approximation to estimate the probability that at least 35 people make a donation.

Hint: Model the number of donations as binomial, then approximate with a normal distribution and use a continuity correction.

Solution 4.43

Let X be the number of donations. Then

$$X \sim \text{Bin}(100, 0.31).$$

So

$$\mu = np = 31, \quad \sigma^2 = np(1-p) = 100(0.31)(0.69) = 21.39.$$

Thus

$$\sigma \approx 4.63.$$

We want

$$P(X \geq 35).$$

Using a continuity correction,

$$P(X \geq 35) \approx P(Y \geq 34.5),$$

where

$$Y \sim N(31, 21.39).$$

Then

$$P(X \geq 35) \approx P\left(Z \geq \frac{34.5 - 31}{4.63}\right) = P(Z \geq 0.76).$$

Hence

$$P(X \geq 35) \approx 1 - 0.7764 = 0.2236.$$

So the required probability is approximately

$$\boxed{0.22}.$$

Takeaways 4.43

- A normal approximation needs the binomial mean and variance first.
- Continuity correction is part of the model, not an optional decoration.

Problem 4.44: Designing the “Lucky 3” lottery game

A local club is designing a new lottery game called *Lucky 3*. A player selects 3 distinct numbers from

$$\{1, 2, 3, 4, 5, 6, 7, 8, 9\},$$

and the host then draws 3 distinct winning numbers from the same set.

Tier 1 pays when all 3 numbers match, in any order. Tier 2 pays when exactly 2 numbers match.

- (i) Show that the total number of possible selections is 84.
- (ii) Let X be the number of matches. Find $P(X = 3)$ and show that

$$P(X = 2) = \frac{3}{14}.$$

- (iii) If the prizes are \$40 for 3 matches, \$10 for exactly 2 matches, and \$0 otherwise, find the expected payout for a single game.
- (iv) If the ticket price is \$5, find the expected profit per ticket for the club and the expected total profit on 1000 tickets.
- (v) Explain why increasing the Tier 2 prize from \$10 to \$15 could be risky for the club, even if the game still has positive expected profit.

Hint: Use combinations for the matching probabilities, then separate *expected payout* from *expected profit*. For the final part, think about how a smaller margin affects risk.

Solution 4.44

(i) The total number of selections is

$$\binom{9}{3} = 84.$$

(ii) For 3 matches,

$$P(X = 3) = \frac{1}{84}.$$

For exactly 2 matches,

$$P(X = 2) = \frac{\binom{3}{2}\binom{6}{1}}{\binom{9}{3}} = \frac{18}{84} = \frac{3}{14}.$$

(iii) If P is the payout, then

$$E(P) = 40 \left(\frac{1}{84} \right) + 10 \left(\frac{3}{14} \right) = \frac{220}{84} = \frac{55}{21} \approx \$2.62.$$

(iv) Expected profit per ticket:

$$5 - \frac{55}{21} = \frac{50}{21} \approx \$2.38.$$

For 1000 tickets, the expected total profit is

$$1000 \cdot \frac{50}{21} \approx \$2381.$$

(v) If the Tier 2 prize becomes \$15, then

$$E(P) = 40 \left(\frac{1}{84} \right) + 15 \left(\frac{3}{14} \right) = \frac{310}{84} \approx \$3.69.$$

So the expected profit per ticket drops to about

$$5 - 3.69 = \$1.31.$$

The game still favours the club on average, but the profit margin is much smaller, so a run of extra Tier 2 winners would create more financial pressure. The game becomes riskier even though the mean profit remains positive.

Takeaways 4.44

- Expected payout and expected profit are different quantities and should be kept separate.
- A positive mean profit does not remove short-run financial risk when prizes are volatile.

Problem 4.45: The truncated St Petersburg game

A fair coin is tossed until a head appears. If the first head appears on toss number k , the prize is

$$W = 2^k \text{ dollars.}$$

The casino caps the game at n tosses. If no head has appeared by the n th toss, the game ends and the player is paid the maximum prize

$$2^n.$$

- (i) Suppose $n = 10$.
 - (a) List the possible values of W and their probabilities.
 - (b) Show that the expected payout is \$11.
 - (c) If the entry fee is \$40, find the casino's expected profit per game.
- (ii) Let E_n be the expected payout for a game capped at n tosses. Show that $E_n = n+1$, then find the value of n for which the expected payout is exactly \$40 and the corresponding maximum jackpot in scientific notation.
- (iii) Using the version with expected payout \$40:
 - (a) Find the probability that a player wins more than the \$40 entry fee.
 - (b) Comment on the difference between expected value and the most likely outcome for the player.

Hint: For the capped game, the first $n-1$ payouts have the usual probabilities $\left(\frac{1}{2}\right)^k$, but the final payout 2^n includes both "first head on toss n " and "no head in the first n tosses".

Solution 4.45

(i) For $n = 10$, the payouts are $2, 4, 8, \dots, 2^{10}$. For $k = 1, \dots, 9$,

$$P(W = 2^k) = \left(\frac{1}{2}\right)^k, \quad P(W = 2^{10}) = \left(\frac{1}{2}\right)^9.$$

Hence

$$E(W) = \sum_{k=1}^9 2^k \left(\frac{1}{2}\right)^k + 2^{10} \left(\frac{1}{2}\right)^9 = 9 + 2 = 11,$$

so the casino's expected profit at a \$40 entry fee is

$$40 - 11 = \$29.$$

(ii) In general,

$$E_n = \sum_{k=1}^{n-1} 2^k \left(\frac{1}{2}\right)^k + 2^n \left(\frac{1}{2}\right)^{n-1} = (n-1) + 2 = n + 1.$$

So $E_n = 40$ gives

$$n + 1 = 40 \implies n = 39.$$

The maximum jackpot is

$$2^{39} = 549755813888 \approx 5.50 \times 10^{11} \text{ dollars.}$$

(iii) To win more than \$40, the player needs $2^k > 40$, so $k \geq 6$. Therefore

$$P(W > 40) = P(\text{no head in the first 5 tosses}) = \left(\frac{1}{2}\right)^5 = \frac{1}{32} \approx 3.1\%.$$

So even when the expected payout is \$40, the player wins more than the entry fee only about 3.1% of the time; the expectation is supported by a tiny chance of a huge jackpot, while the most likely practical outcome is a loss.

Takeaways 4.45

- Expected value can be dominated by very rare, very large outcomes.
- This is a classic reminder that “fair on average” and “likely to feel fair” are not the same idea.

Problem 4.46: HSC 2025-style: Inverse-tail green-counter problem

A bag contains counters, some of which are green. One hundred trials are run. In each trial, one counter is selected at random, its colour is noted, and the counter is returned to the bag.

Let X be the number of times that a green counter is selected. It is known that

$$E(X) = 20$$

and

$$P(X \geq k) = 0.0668.$$

Find the value of k using a normal approximation.

Hint: From $E(X) = 20$ with 100 trials, find p , then approximate the binomial distribution by a normal distribution and work backwards from the tail probability.

Solution 4.46

Since

$$E(X) = np = 100p = 20,$$

we have

$$p = 0.2.$$

So

$$X \sim \text{Bin}(100, 0.2).$$

Hence

$$\mu = np = 20, \quad \sigma^2 = np(1-p) = 100(0.2)(0.8) = 16,$$

so

$$\sigma = 4.$$

Now

$$P(X \geq k) = 0.0668.$$

Using the standard normal distribution, this corresponds to an upper-tail z -value of about 1.5, since

$$P(Z \geq 1.5) \approx 0.0668.$$

Using a continuity correction,

$$\frac{k - 0.5 - 20}{4} \approx 1.5.$$

So

$$k - 0.5 \approx 26$$

and therefore

$$k \approx 26.5.$$

Hence the required integer value is

$$\boxed{27}.$$

Takeaways 4.46

- In inverse-tail questions, work backward from the normal tail to a z -score.
- Continuity correction is important when converting the binomial threshold to a normal one.

Problem 4.47: HSC 2022-style: Airline overbooking threshold

An airline has 350 seats on a flight. Historical data suggests that each booked passenger independently fails to show up with probability 0.05.

The airline wants to sell the largest number N of tickets such that the probability of more passengers arriving than there are seats is less than 0.01.

Explain how this can be modelled, and identify the inequality that N must satisfy.

Hint: Model the number of passengers who arrive, or equivalently the number who do not show up.

Solution 4.47

Let X be the number of passengers who arrive. Then each passenger arrives with probability 0.95,

so

$$X \sim \text{Bin}(N, 0.95).$$

If more passengers arrive than there are seats, then

$$X > 350.$$

So the airline wants the largest value of N such that

$$P(X > 350) < 0.01.$$

Equivalently, if Y is the number of no-shows, then

$$Y \sim \text{Bin}(N, 0.05)$$

and the condition becomes

$$P(Y < N - 350) < 0.01.$$

This sets up the threshold problem, which can then be solved approximately using a normal model.

Takeaways 4.47

- Modelling is the main goal here: define the random variable and write the correct threshold inequality.
- Overbooking questions connect expectation ideas to decision-making under risk.

5 Appendices

5.1 Appendix A: Formula Sheet

This appendix is designed as a last-minute revision sheet. The formulas matter, but the real skill is knowing when each one applies.

Event Probability

If all outcomes are equally likely, then

$$P(A) = \frac{n(A)}{n(S)}.$$

Complement Rule

$$P(A^c) = 1 - P(A).$$

Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A and B are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B).$$

Conditional Probability

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

Multiplication Rule

$$P(A \cap B) = P(A)P(B | A) = P(B)P(A | B).$$

Independence

If A and B are independent, then

$$P(A \cap B) = P(A)P(B).$$

Expectation and Variance

For a discrete random variable X ,

$$E(X) = \sum xP(X = x),$$

$$E(X^2) = \sum x^2P(X = x),$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2.$$

Binomial Distribution

If

$$X \sim \text{Bin}(n, p),$$

then

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}, \quad r = 0, 1, \dots, n,$$

$$E(X) = np, \quad \text{Var}(X) = np(1-p).$$

Counting Formulas

$${}_nP_r = \frac{n!}{(n-r)!}, \quad \binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

When to Use What

Question wording	Likely method
at least one	complement rule
given that	conditional probability
without replacement	multiplication rule with changing fractions
committee or selection	combinations
exactly r successes in repeated trials	binomial formula
fair game or average return	expected value

5.2 Appendix B: Common Probability Traps

- **Mixing up mutually exclusive and independent.** Mutually exclusive events cannot happen together. Independent events do not affect each other's probabilities. These are very different ideas.
- **Forgetting to subtract the overlap in “or” questions.** If two events can happen together, the addition rule needs the intersection subtracted once.
- **Using constant probabilities without replacement.** Once an object is removed, the denominator usually changes.
- **Calling a question binomial too early.** Check all four ingredients: fixed number of trials, two outcomes per trial, constant success probability, and independence.
- **Confusing expectation with a guaranteed outcome.** The expected value is a long-run average, not a promise for a single trial.
- **Using the wrong success probability in a binomial question.** The value of p must match the event you are counting as a success.
- **Dropping the combination factor in binomial probability.** The term $\binom{n}{r}$ counts how many different positions the successes can occupy.
- **Using ordered counting when order does not matter.** In committee-style questions, combinations usually keep the probability model clean.

- **Forgetting that probability tables must sum to 1.** When one probability is unknown, always use this fact first.
- **Using a reduced sample space incorrectly in conditional probability.** Once the condition is given, the denominator should reflect that condition only.

A strong exam habit is to write one short line naming the method before calculating. For example:

- “Use complement since ‘at least one’ is awkward directly.”
- “Use conditional probability because the sample space has changed.”
- “This is binomial with $n = 5$ and $p = 0.3$.”

That brief method statement often prevents a correct formula from being used in the wrong setting.

6 Conclusion

Probability in Extension 1 is about much more than substituting into formulas. Strong students learn to identify the sample space carefully, translate words into events, and choose the right model for repeated trials or random variables.

The key habits to build are:

- define events clearly before calculating
- use complements whenever direct counting is messy
- distinguish conditional probability from ordinary probability
- treat expectation as a weighted average, not a guaranteed outcome
- recognise binomial structure only when the repeated-trial conditions really hold

With steady practice, probability becomes one of the most readable parts of the course. Best of luck with your Extension 1 studies and HSC preparation.

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